TRAFFIC LIGHT CONTROLLER WITH PEDESTRIAN PRIORITY

WINTER INTERNSHIP REPORT : DECEMBER 2025

INTERNSHIP REPORT SUBMITTED IN FULFILMENT OF THE REQUIREMENTS FOR ASSESSMENT AND AWARD OF CREDITS FOR THE COURSE UNDER THE AUTONOMOUS REGULATIONS OF SRI ESHWAR COLLEGE OF ENGINEERING, COIMBATORE AFFILIATED TO ANNA UNIVERSITY, CHENNAI

INTERNSHIP REPORT

Submitted by

LAKSANA A-722824106077

BATCH 2024 – 2028

Under the Guidance of

SREEJEESH G

SENIOR TECHNICAL OFFICIER

VLSI GROUP

NIELIT,CALICUT

Sri Eshwar College of Engineering

(An Autonomous Institution)

Kinathukadavu (Tk), Coimbatore - 641 202, Tamil Nadu

Approved by AICTE, New Delhi and Affiliated to Anna University, Chennai

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PROGRESS THROUGH KNOWLEDGE

INTERNSHIP REPORT

Submitted by

LOGADHARSHINI S-722824106080

LAKSANA A-722824106077

BATCH 2024 – 2028

Under the Guidance of

SREEJEESH G

SENIOR TECHNICAL OFFICIER

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NIELIT,CALICUT

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that the internship project entitled **TRAFFIC LIGHT CONTROLLER WITH PEDESTRIAN PRIORITY** was carried out byLaksana A,722824106117 during the short-term internship programme, in partial fulfillment of the requirements for the award of the Internship Certificate.

The work presented in this Internship Project Report is a record of original work carried out by the student under my supervision and guidance in the Department of VLSI Group NIELIT,Calicut

To the best of my knowledge, the work is genuine and has not been submitted elsewhere for any other academic or professional purpose.

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SIGNATURE	**SIGNATURE**
SREEJEESH G **SENIOR TECHNICAL OFFICIER** **DEPARTMENT OF VLSI** **NIELIT, CALICUT**	**SREEJEESH G** **SENIOR TECHNICAL OFFICIER** **DEPARTMENT OF VLSI** **NIELIT, CALICUT**

Submitted for the **as a part of winter Internship** held during the period from 01.12.2025 to 26.12.2025

CANDIDATE'S DECLARATION

.

I, LAKSANA A hereby declare that the report entitled "DIGITAL PROTOTYPING USING FGPA" which is being submitted to DEPARTMENT OF VLSI in **NIELIT ,CALICUT**, is our authentic work carried out during internship.

I declare that my work has not been submitted in part or in full to any other university or institution for the award of any certificate.

LAKSANA A

QUALITY POLICY

To establish a system of Quality Enhancement, which would on a continuous basis evaluate and enhance the quality of teaching – learning, research and extension activities of the institution, leading to improvements in all processes, enabling the institution to attain excellence.

INSTITUTE VISION

To be recognized as a premier institution, grooming students into globally acknowledged engineering professionals.

INSTITUTE MISSION

- Providing outcome and value-based engineering education
- Nurturing research and entrepreneurial culture
- Enabling students to be industry ready and fulfill their career aspirations
- Grooming students through behavioral and leadership training programs
- Making students socially responsible

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DEPARTMENT VISION

To be recognized as a premier institution, grooming students into globally acknowledged engineering professionals

PROGRAM EDUCATIONAL OBJECTIVES

- PEO1: Pursue career in multinational organizations, research organizations and core industries, higher studies at premier institutions and establish start-ups.
- PEO2: Acquire core competencies in Electronics and Communication Engineering and exposure to latest Electronic Design Automation (EDA) tools.
- PEO3: Exhibit professional skills and collaborative work experience

PROGRAM OUTCOMES

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one’s own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES

PSO1:Interpret and design electronic systems using internet of things, VLSI technology and efficient signal processing and computing algorithms.

PSO2:Apply knowledge to solve challenges in communication systems and networks.

INTERNSHIP PROJECT REPORT

Executive Summary

This internship project at NIELIT Calicut involves the design and FPGA implementation of a Traffic Light Controller with Pedestrian Priority, aimed at developing a reliable and real-time digital control system for road safety and traffic management. The project's objective was to create an automated traffic signal mechanism that follows a fixed sequence of Green, Yellow, and Red lights while intelligently responding to pedestrian requests and emergency conditions. The controller uses a dedicated pedestrian push button, which immediately prioritizes pedestrian safety by transitioning the signal to Red and activating a countdown timer on the seven-segment display, allowing safe crossing within a defined time window. Additionally, an emergency button overrides all normal operations, ensuring that the traffic lights switch instantly to Red during critical situations. The methodology followed a structured digital design flow that included requirement analysis, FSM design, timing diagram creation, module-level Verilog implementation, simulation using WaveADO/Vivado, and hardware realization on the Basys 3 FPGA board. Essential components such as clock dividers, timers, debouncing circuits, and display drivers were implemented to ensure accurate timing and reliable input handling. Tools like Vivado Design Suite and the WaveADO simulator were used extensively for verifying system functionality before deployment. Hardware testing confirmed that all modules worked consistently, with correct state transitions, stable timing, and accurate seven-segment countdown output. The project provided significant hands-on experience in FPGA prototyping, Verilog HDL coding, synchronous digital design, finite state machine implementation, and hardware debugging, strengthening key skills relevant to the VLSI domain. Overall, this internship project successfully delivered a functional and safety-oriented traffic control system that demonstrates the effectiveness of FPGA-based real-time design while greatly enhancing practical learning and understanding of digital system development.

Table of Content

CHAPTER 1: INTRODUCTION

1.1 Organization Background

NIELIT Calicut is a premier institute specializing in advanced training, research, and consultancy in the areas of VLSI, embedded systems, digital design, and FPGA development. The institute provides hands-on exposure to industry-oriented technologies and tools. As part of the internship, students work on real-time hardware projects that strengthen practical understanding of digital design concepts**.**

1.2 Objectives of the Internship

- To gain practical knowledge of digital system design on FPGA platforms.
- To understand the development flow using Verilog HDL and hardware tools like Vivado.
- To design and implement a real-time traffic control system with pedestrian priority.
- To apply finite state machine (FSM) concepts for sequencing and timing control.
- To develop skills in simulation, synthesis, and hardware testing on the Basys- 3 fpga board.

1.3 Scope of the Internship

The internship focuses on designing a complete digital system—from concept to FPGA implementation. The project includes:

- Real-time traffic light management.
- Pedestrian safety using a priority button.
- Emergency override using an emergency switch.
- Seven-segment countdown display.
- Timing control using clock dividers and FSM.
- Hardware validation on the Basys 3 FPGA board.

CHAPTER 2: INTERNSHIP ACTIVITIES AND PROJECT WORK

2.1 Roles and Responsibilities

- Understanding the problem statement and system requirements.
- Designing the block diagram and FSM architecture.
- Writing Verilog modules for the controller, timers, and display.
- Performing simulation and debugging.
- Creating pin constraints for Basys 3 board.
- Generating bitstream and testing on hardware.
- Documenting project workflow and results.

2.2 Project Description / Modules Undertaken

The project is a **Traffic Light Controller with Pedestrian Priority**, designed using Verilog HDL. It consists of:

- Traffic signal control module
- Pedestrian button interface
- Emergency override module
- Countdown display controller
- Timing control module
- Top-level integration module

2.3 Methodology / Algorithm / System Design

- **Requirement Analysis:** Identified traffic flow rules, light timing, and pedestrian needs.
- **FSM Design:** Developed state diagrams for normal operation, pedestrian request states, and emergency state.
- **Clock Divider:** Generated 1 Hz pulses from FPGA’s 100 MHz clock for real-time timing.
- **Module Development:** Implemented separate Verilog modules for signals, counter, and display.
- **Integration:** Connected all sub-modules in a top-level design.
- **Simulation:** Verified all timing and signal transitions.
- **Hardware Implementation:** Synthesized, implemented, and tested on FPGA.

2.4 Tools and Technologies Used

- **Hardware:** Basys 3 FPGA Board.
- **Software:** Vivado Design Suite / WaveADO Simulator.
- **Language:** Verilog HDL.
- **Hardware Accessories:** LEDs, push buttons, seven-segment display.

2.5 Results and Outcome Analysis

- Normal traffic cycle (Green → Yellow → Red) implemented with correct timing.
- Pedestrian button successfully gives priority by forcing stop (Red).
- Seven-segment display shows **countdown timer** during pedestrian crossing.
- Emergency button overrides all signals, forcing immediate Red for safety.
- Successful hardware testing confirming real-time operation.

CHAPTER 3: SKILLS AND LEARNING OUTCOMES

3.1 Technical Skills Acquired

- Verilog HDL coding and debugging.
- FSM-based digital system design.
- Seven-segment display interfacing.
- Clock dividers and timing management.
- Simulation techniques using WaveADO/Vivado.
- FPGA implementation, pin mapping, and hardware testing.

3.2 Professional Skills Acquired

- Project documentation and version control.
- Time management and systematic debugging approach.
- Team coordination and technical communication.
- Understanding real-world constraints in digital systems.

3.3 Personal and Career Development

- Developed a systematic problem-solving approach by designing and implementing an FSM-based traffic controller.
- Improved logical thinking, timing analysis, and debugging skills through simulation and waveform verification.
- Gained hands-on experience with Verilog HDL and the complete FPGA design flow, from coding to implementation.
- Strengthened self-learning, patience, and attention to detail while resolving design and simulation issues.
- Enhanced technical communication skills by explaining design logic, state transitions, and results clearly.

CHAPTER 4: CHALLENGES AND SOLUTIONS

4.1 Challenges Faced

- **Incorrect timing generation causing unstable state transitions**
  Initially, improper clock timing led to irregular state changes in the FSM, resulting in signals switching too fast or skipping expected states. This made the traffic sequence unreliable during simulation and hardware testing.
- **Synchronizing pedestrian button input with FSM**
  The pedestrian request button was asynchronous with respect to the system clock, which caused multiple unintended triggers and unpredictable FSM behavior when the button was pressed.
- **Mapping I/O pins of Basys 3 without conflicts**
  Assigning incorrect or overlapping pin constraints led to synthesis and implementation issues, preventing proper interaction between input buttons, LEDs, and display modules on the FPGA board.
- **Seven-segment display not showing correct countdown initially**
  The countdown timer values were not displayed correctly due to incorrect BCD conversion and improper digit multiplexing, causing flickering and wrong numerical outputs.
- **Emergency module affecting normal state execution**
  The emergency override logic interfered with the normal FSM flow, sometimes blocking standard traffic sequences and preventing the system from returning to its default operating mode.

4.2 Solutions and Strategies Adopted

- **Used clock divider for stable 1 Hz pulses**
  A clock divider module was implemented to derive a precise 1 Hz timing signal from the FPGA's high-frequency clock, ensuring stable and predictable state transitions in the FSM.
- **Debounced pedestrian button input to avoid glitches**
  A debouncing mechanism was introduced to filter out mechanical noise from the pedestrian button, ensuring that each press generated a single, clean request signal synchronized with the system clock.
- **Rechecked constraint file and used proper FPGA pin assignments**
  The constraint (.xdc) file was carefully reviewed and updated according to the Basys 3 reference manual, eliminating pin conflicts and ensuring correct mapping of inputs and outputs.
- **Designed BCD counters with multiplexed display driver**
  BCD counters were implemented along with a time-multiplexed seven-segment display driver to accurately show countdown values while minimizing hardware resource usage.
- **Isolated emergency logic with highest state priority in FSM**
  The emergency functionality was redesigned as a high-priority state within the FSM, allowing immediate override when required while ensuring a smooth and safe return to normal traffic operation.

CHAPTER 5: CONCLUSION AND FUTURE

Conclusion:

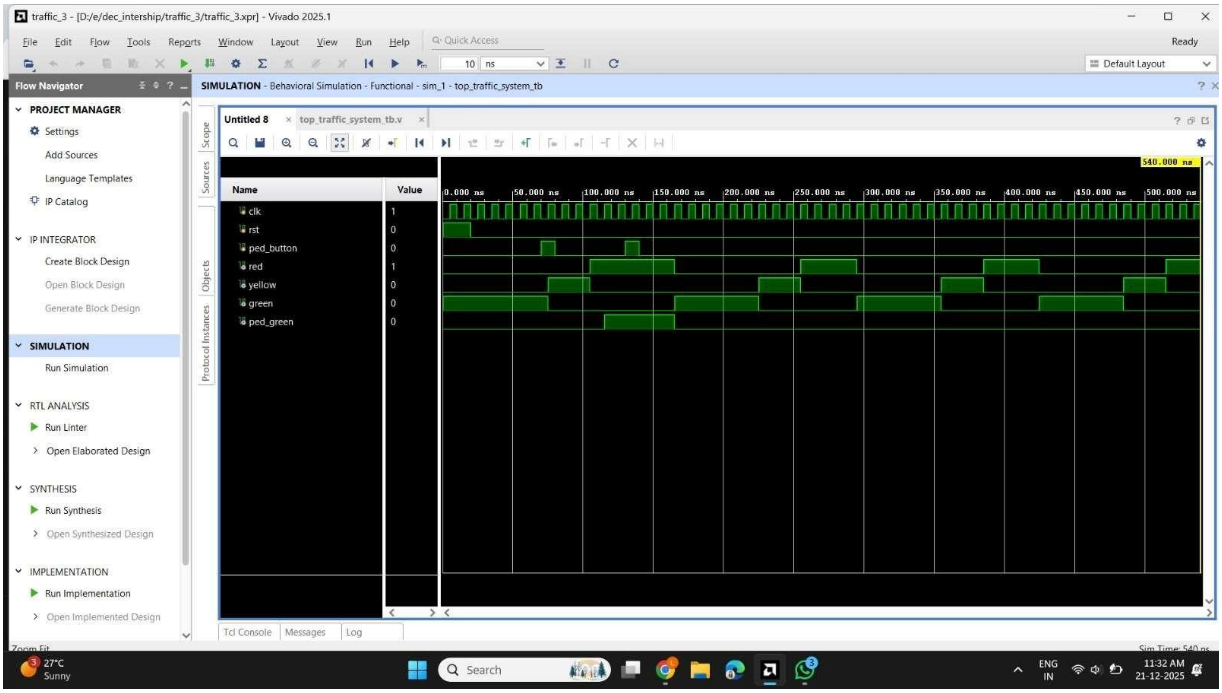
The project **"Traffic Light Controller with Pedestrian Priority Using FPGA"** was successfully developed and implemented as part of the internship at NIELIT Calicut, fulfilling all functional and design objectives. The system automates the standard Red–Yellow–Green traffic sequence based on precise timing control and provides enhanced safety by giving priority to pedestrians whenever the push button is pressed, immediately switching to a Red signal and displaying a countdown on the seven-segment display. An emergency button was also incorporated to override normal operation and ensure immediate stopping of traffic during critical situations. The design was implemented using Verilog HDL and FSM-based logic, and validated through simulation, synthesis, and hardware testing on the Basys 3 FPGA board. This project helped in understanding real-time digital design, clock division, debouncing techniques, and modular hardware architecture. Overall, the work strengthened practical skills in FPGA prototyping and provided valuable exposure to VLSI concepts, digital system development, and real-world problem solving in intelligent traffic control systems.

Future Scope:

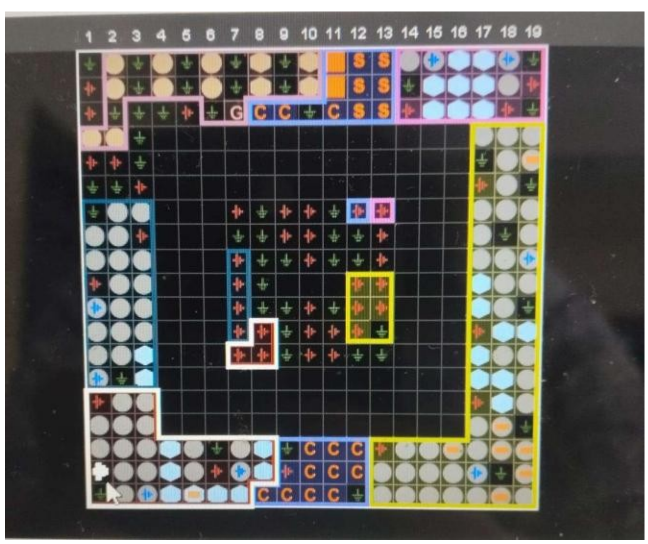
- Adding vehicle sensors for adaptive traffic control.
- Wireless communication between traffic junctions.
- Integration of IoT-based monitoring.
- Displaying additional information on LCD or OLED screens.
- Implementing machine learning–based traffic prediction.

CHAPTER 6: APPENDICES

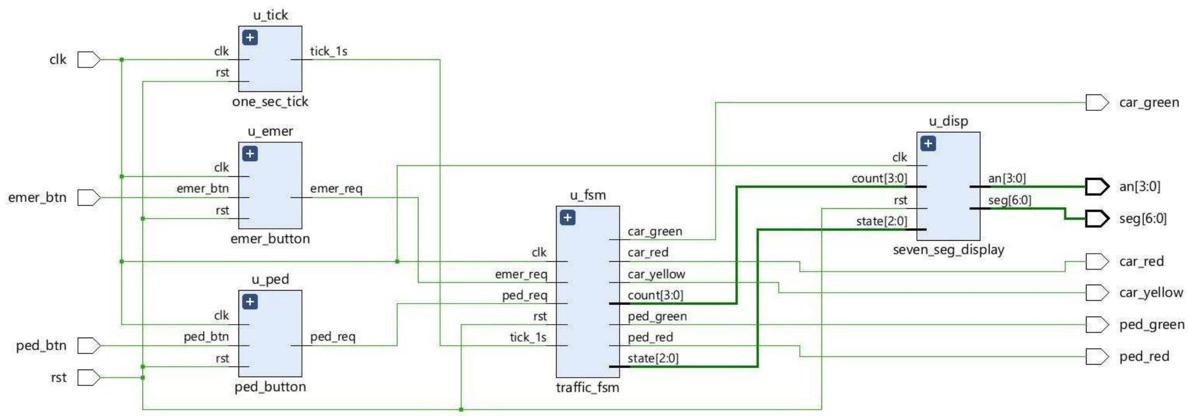
6.1 Simulation Waveforms



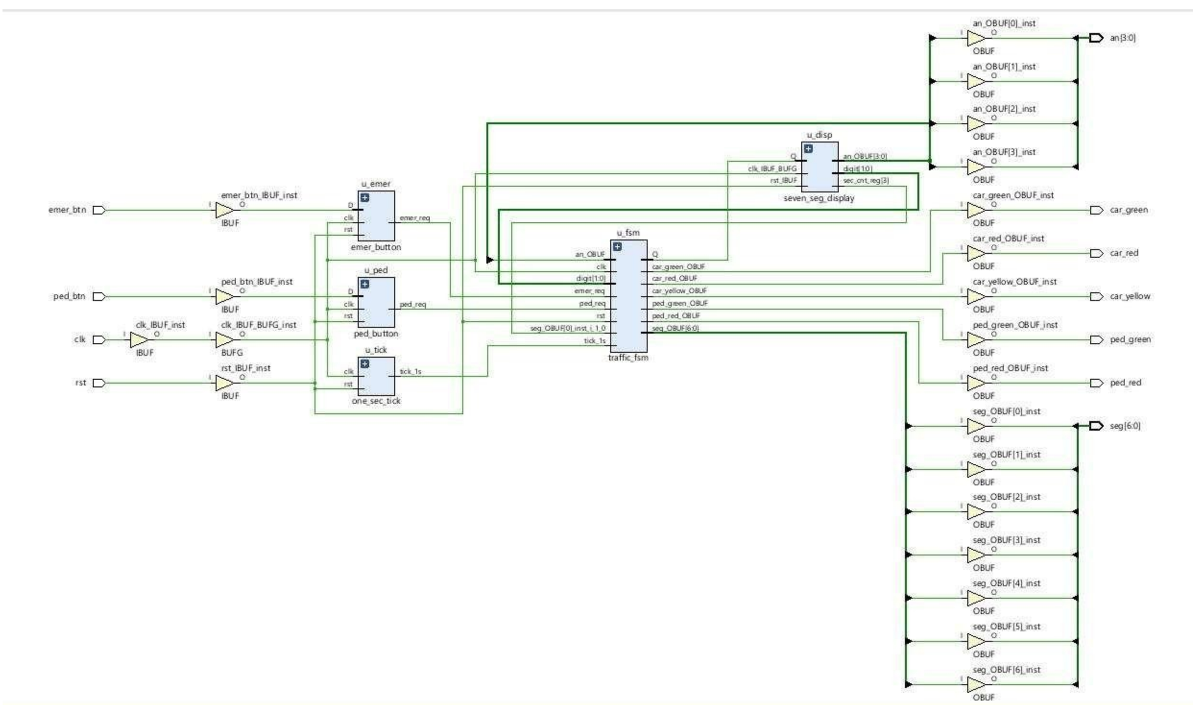
OUTPUT WAVEFORM



6.2 Block Diagram and FSM Diagram



RTL ANALYSIS



RTL SYNTHESIS

6.3 Verilog Code

```
module traffic_light_top (
    input  wire clk,
    input  wire rst,
    input  wire ped_btn,
    input  wire emer_btn,
    output wire car_red,
    output wire car_yellow,
    output wire car_green,
    output wire ped_red,
    output wire ped_green,
    output wire [6:0] seg,
    output wire [3:0] an
);
    wire tick_1s;
    wire ped_req;
    wire emer_req;
    wire [2:0] state;
    wire [3:0] count;
    one_sec_tick u_tick (
        .clk(clk),
        .rst(rst),
        .tick_1s(tick_1s)
    );
    ped_button u_ped (
        .clk(clk),
        .rst(rst),
        .ped_btn(ped_btn),
        .ped_req(ped_req)
    );
    emer_button u_emer (
```

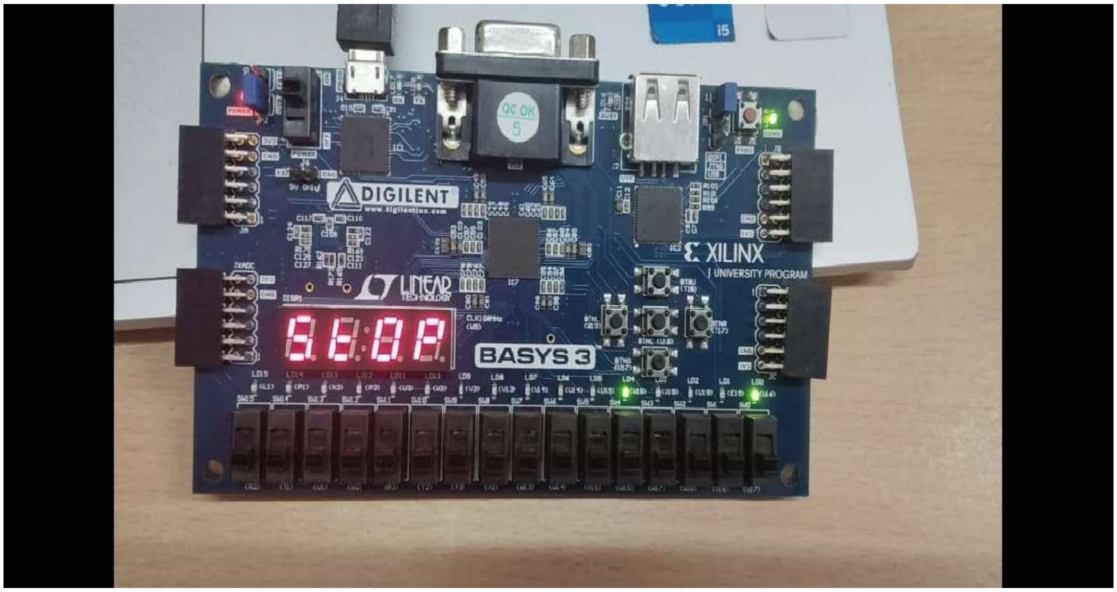
```
        .clk(clk),
        .rst(rst),
        .emer_btn(emer_btn),
        .emer_req(emer_req)
    );

    traffic_fsm u_fsm (
        .clk(clk),
        .rst(rst),
        .tick_1s(tick_1s),
        .ped_req(ped_req),
        .emer_req(emer_req),
        .state(state),
        .count(count),
        .car_red(car_red),
        .car_yellow(car_yellow),
        .car_green(car_green),
        .ped_red(ped_red),
        .ped_green(ped_green)
    );

    seven_seg_display u_disp (
        .clk(clk),
        .rst(rst),
        .state(state),
        .count(count),
        .seg(seg),
        .an(an)
    );
endmodule
```

6.4 Pin constraint file (XDC)

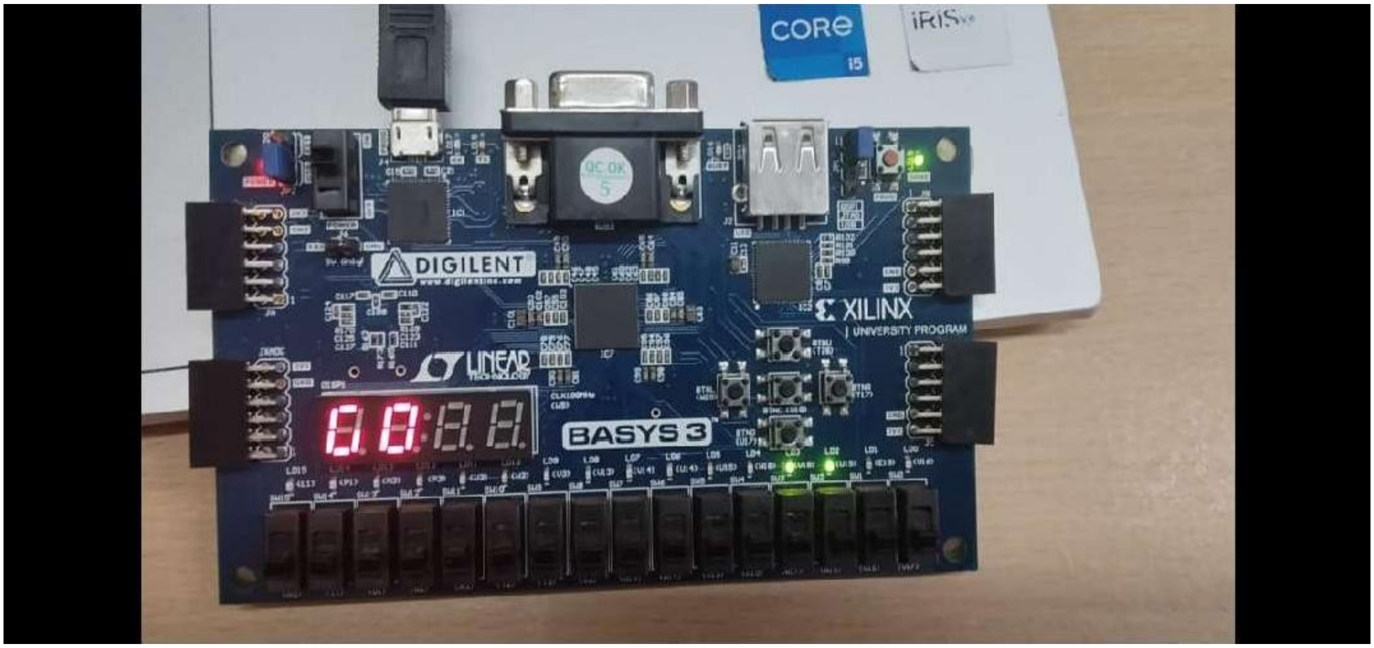
```
############################################################
## Basys 3 Constraint File
## FPGA  : XC7A35T-1CPG236C
## Clock : 100 MHz
###########################################################
############################
## CLOCK INPUT
###########################
set_property PACKAGE_PIN W5 [get_ports clk]
set_property IOSTANDARD LVCMOS33 [get_ports clk]
create_clock -name sys_clk -period 10.000 [get_ports clk]

###########################
## PUSH BUTTON INPUTS
###########################

## Reset button (btnC)
set_property PACKAGE_PIN U18 [get_ports rst]
set_property IOSTANDARD LVCMOS33 [get_ports rst]

## Pedestrian button (btnU)
set_property PACKAGE_PIN T18 [get_ports ped_btn]
set_property IOSTANDARD LVCMOS33 [get_ports ped_btn]

## Emergency button (btnL)
set_property PACKAGE_PIN W19 [get_ports emer_btn]
set_property IOSTANDARD LVCMOS33 [get_ports emer_btn]
############################
## LED OUTPUTS
###########################

## Car traffic LEDs
```

```
set_property PACKAGE_PIN U16 [get_ports car_red]       ;# LD0
set_property PACKAGE_PIN E19 [get_ports car_yellow]  ;# LD1
set_property PACKAGE_PIN U19 [get_ports car_green]     ;# LD2
## Pedestrian LEDs
set_property PACKAGE_PIN V19 [get_ports ped_red]       ;# LD3
set_property PACKAGE_PIN W18 [get_ports ped_green]     ;# LD4
set_property IOSTANDARD LVCMOS33 [get_ports {car_red car_yellow car_green ped_red ped_green}]
#############################
## SEVEN-SEGMENT DISPLAY
## Common Anode, Active LOW
############################
## Segment pins (a b c d e f g)
set_property PACKAGE_PIN W7 [get_ports seg[0]]    ;# a
set_property PACKAGE_PIN W6 [get_ports seg[1]]  ;# b
set_property PACKAGE_PIN U8 [get_ports seg[2]]     ;# c
set_property PACKAGE_PIN V8 [get_ports seg[3]]     ;# d
set_property PACKAGE_PIN U5 [get_ports seg[4]]     ;# e
set_property PACKAGE_PIN V5 [get_ports seg[5]]     ;# f
set_property PACKAGE_PIN U7 [get_ports seg[6]]     ;# g

## Digit enable (anodes)
set_property PACKAGE_PIN U2 [get_ports an[0]]     ;# rightmost digit
set_property PACKAGE_PIN U4 [get_ports an[1]]
set_property PACKAGE_PIN V4 [get_ports an[2]]
set_property PACKAGE_PIN W4 [get_ports an[3]]       ;# leftmost digit

## Correct wildcard usage
set_property  IOSTANDARD  LVCMOS33 [get_ports {seg[*] an[*]}]

##########################################################
## END OF CONSTRAINT FILE
##########################################################
```

6.5 HARDWARE TESTING PHOTOS



The traffic light controller is designed to operate automatically by cycling the three traffic signals—Green, Yellow, and Red—in a continuous loop, with each signal active for a fixed duration such as five seconds for Green and Red and three seconds for Yellow. Under normal operation, the lights transition smoothly in the sequence Green → Yellow → Red and repeat, ensuring regulated traffic flow. When the pedestrian button is pressed, the controller immediately overrides the normal cycle and forces the traffic light to switch to Red, while activating the pedestrian Green signal to allow safe crossing. In emergency situations, pressing the emergency button causes the system to break out of all normal and pedestrian modes and enter an emergency state where the Red light begins blinking continuously as a warning signal. Along with the light control, the system also interfaces with a seven-segment display that shows status messages such as “GO” during Green, “WAIT” during Yellow, “STOP” when Red is active, and “EMG” or a similar indicator during emergency mode. This integrated design ensures smooth traffic flow, safe pedestrian crossing, and immediate response during critical situations, making the controller efficient and reliable for real-time traffic management.



The image shows the successful hardware implementation of a traffic control system with pedestrian safety on the BASYS-3 FPGA board. The FPGA is powered and programmed, with on-board LEDs representing vehicle traffic signals such as red, yellow, and green, while a separate LED indicates the pedestrian crossing status. The 7-segment display is active and shows the timing or state information, confirming that the clock divider and control logic are functioning correctly. The orderly LED transitions demonstrate that the finite state machine is operating as intended, ensuring safe and sequential control of vehicle and pedestrian movements.

7: REFERENCES

General FPGA and Digital Design

1. Xilinx Vivado Design Suite Documentation
   https://www.xilinx.com/support/documentation/sw_manuals/xilinx2020_2/
2. FPGA Tutorials and Basics – GeeksforGeeks
   https://www.geeksforgeeks.org/fpga-field-programmable-gate-array/
3. Introduction to Digital Logic and FSMs – TutorialsPoint
   https://www.tutorialspoint.com/digital_logic/digital_fsm.htm
4. Verilog HDL Tutorial – Asic World
   http://www.asic-world.com/verilog/
5. Verilog for FPGA Design – FPGA4student
   https://www.fpga4student.com/p/verilog.html
6. Verilog HDL Reference – EDA Playground
   https://www.edaplayground.com/

PROJECT TITLE	**TRAFFIC LIGHT CONTROLLER WITH PEDESTRIAN PRIORITY**	
PROGRAM	**NIELIT INTERNSHIP PROGRAM**	
PROJECT BATCH NUMBER	**2**	
BATCH MEMBERS	**LOGADHARSHINI S**	**722824106080**
	LAKSANA A	**722824106077**
NAME OF THE SUPERVISOR	**SREEJESH G**	
NAME OF THE SDG GOALS MAPPED	**Sustainable Cities and Communities, Good Health and Well-Being, Sustainable Cities and Communities**	
MENTION THE SDG GOALS NUMBER	**SDG-9, SDG-3, SDG-11**	
NAME OF THE TRL LEVEL	**Prototype Demonstrated in a Relevant Environment**	
MENTION THE TRL LEVEL	**TRL LEVEL-6**	

POs & PSOs Mapping (Put a tick mark in the mapped PO's & PSO's):

Program Outcomes											**Program Specific Outcomes**		
PO1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PSO 1**	**PSO 2**	**PSO 3**

Signature of the Mentor

VENUE AND EXPENDITURE STATEMENT FOR THE PROJECT WORK

Laboratory details where the project is carried out	**NIELIT**
Software / Hardware details	**VIVADO SOFWARE,BASYS-3 FPGA**

Signature of the student **Signature of the Mentor**

(Names of the Student) **(Name of the Supervisor)**